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A recursive system identification inertia estimator for traditional and converter-interfaced generators

Davide Gotti *, Pablo Ledesma, Hortensia Amaris

Universidad Carlos III de Madrid, Av. Universidad 30, Leganés, 28911, Madrid, Spain

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ABSTRACT

This paper proposes a method to estimate the inertia constant of synchronous generators (SGs) and the virtual inertia provided by converter-interfaced generators (CIGs). It can be applied when the system is under normal operating conditions or after power imbalance events. The algorithm is based on a recursive system identification approach and uses the active power and bus frequency measurements proceeding from phasor measurement units. The measurements are preprocessed to improve the estimation accuracy. Subsequently, a model-independent rotor speed estimator is used to compute rotor speed variations (or internal frequency variations, in the case of CIGs). Afterwards, a recursive least-square equation error formulation is implemented to compute the SG and CIG parameters in the z-domain, which are later converted into their respective counterparts in the s-domain. The method is applied to several generation devices in the IEEE 39-bus and IEEE 118-bus test systems, including traditional SGs and CIGs providing virtual inertia. Numerical results show the accuracy and the low computational burden of the proposed method.

1. Introduction

Motivation

In traditional power systems, rotor masses of synchronous generators (SGs) provide the inertial response, which is the first reaction to power unbalances that limits the rate of change of frequency and the nadir frequency. In this manner, the system operation can be maintained within its operational limits, avoiding the unintended tripping of protection systems. However, over the past decades, converted-interfaced generators (CIGs) have been integrated into power systems, gradually replacing SGs. As this transition takes place, the system inertia reduces, posing unprecedented challenges for power systems operation and control [1].

To safely increase the amount of nonsynchronous generation in power systems, CIGs are required to provide an emulated inertia and other services, such as primary frequency control (PFC), damping, and reactive power control. In fact, several inertia and damping control strategies have been recently proposed for PV plants [2,3] and wind farms [4]. One of the challenges that transmission system operators (TSOs) face is to determine whether the CIGs actually provide these control services. In fact, CIG control parameters may be time-variable, and this could expose the power system to security issues if the control is not provided or is unavailable. This study proposes an online parameter estimator applicable for both traditional SGs and CIGs, with the goal of providing the TSO with a clear picture of the actual behavior of the generation devices.

Literature review of inertia estimation methods

Several power system inertia estimation methods have been proposed in recent years. A comprehensive overview can be found in [5,6].

Some of the inertia estimation methods have an area-wide scope, focusing on estimating the total inertia of a power system. For example, the authors of [7] propose an ARMAX-based algorithm that makes use of ambient measurements to provide inertia estimation of power system areas. To compute the aggregated area frequency of each system, a weight must be assigned to each generator. The assignment of these weights is empirical, and this could lead to estimation errors. Furthermore, the CPU burden is on the order of tens of seconds, which could pose limitations for protection and control applications.

A switching Markov Gaussian model for power system inertia estimation is proposed in [8]. However, the method is valid only during normal operating conditions. If the frequency deviations are significant, estimations may experience large biases. Furthermore, the methodology relies upon a single frequency measurement, and this could be a limitation for large-scale applications since the selection of a "representative" bus may lead to large approximations [1].

* Corresponding author.
E-mail addresses: dgotti@ing.uc3m.es (D. Gotti), pablole@ing.uc3m.es (P. Ledesma), hamaris@ing.uc3m.es (H. Amaris).

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Ref. [9] proposes a dynamic regressor extension and mixing procedure to estimate the system inertia, among other aggregated parameters. The algorithm is tested on the 21382 bus ENTSO-E network, and the outlined results are accurate. However, since CPU times are not reported, it is difficult to assess its online applicability. Moreover, the method has been tested considering only inertia contributions from SGs and mass-equipped loads, whereas a near-future scenario with virtual inertia provided by CIGs is not considered.

Ref. [10] introduces the estimation of the signal parameter via rotational invariance to estimate the system inertia. The algorithm provides good result accuracy considering the presence of both SGs and photovoltaic (PV) sources. The computational time is approximately tens of milliseconds; therefore, the algorithm is suitable for online applications. Nevertheless, the methodology is sensitive to a number of selections, such as proper model order (number of modes), window length, and the selection of the qth coefficients of the discrete Fourier transform of a given window. These factors appear to be critical, but no guidelines about their selection are provided [11].

Other studies have a wide-element scope, focusing on the estimation of single device parameters, such as SGs and CIGs. For instance, a Bayesian approach for SG parameter estimation is proposed in [12]. The reported results are accurate, and the method is also capable of providing the uncertainty of the estimates. However, the computational time ranges from a few tenths of seconds to thousands of seconds, which poses a limitation for online inertia estimation.

A method to estimate the power system inertia immediately after a disturbance is proposed in [13]. The power system inertia is computed as the sum of the individual generator inertia constants. Reported results present good accuracy and a small computational burden. However, its implementation requires the instant of the disturbance and its power imbalance size to be known, which is not always available to the TSOs.

A new index called the rate of change of power (RoCoP) is proposed in [14]. This index can be used, among other applications, to estimate the inertia provided by a single device. It is applicable to both traditional SGs and CIGs, as shown in [15]. However, this formulation is prone to numerical issues.

To mitigate this problem, [16] converts the previous algebraic formulation into a first-order differential equation. A second formulation is also proposed, allowing the determination of the PFC droop gain. The reported results are accurate, the method is applicable to both SGs and CIGs, and the trajectories of the estimation reach the steady-state value fairly quickly. However, the time step used for all the simulations is 1 ms, whereas the phasor measurement units (PMU) reporting rates are on the order of tens of frames per second [17]. Since this method still relies on the RoCoP, it is worth mentioning that this index performs better the higher the branch impedance among frequency measurements is. If a branch impedance is not sufficiently large, the accuracy required to measure the frequency variations might be too high, compromising the method accuracy [15]. It is possible to use measurements at more distant buses, but it may be impractical since it would increase the number of PMUs to install.

A least square method for estimating the inertia of SGs and CIGs is proposed in [18]. The algorithm mitigates the numerical problems of [15] by applying the least square method over an observation window. The required inputs are the RoCoP and the second-time derivative of internal frequencies, which also need to be estimated. Estimation errors are quite low, but the inertia estimation of CIGs is carried out without considering measurement noise and the method has been tested solely on the three-generators IEEE 9-bus system.

A method for online identification of the inertia distribution under normal operating conditions is proposed in [19]. The method uses the frequency divider formula and the SG swing equation to extract plausible bus inertia values from ambient measurements. However, the reported results are not very accurate. An extension of the approach is presented in [20], proposing the application of a numerical algorithm for subspace system identification to estimate the inertia of individual SGs. The estimation accuracy has improved with respect to previous work [19], but no information about the applicability of this method to CIGs is provided. Furthermore, this approach seems reliable only under normal conditions since it uses the generation bus frequencies instead of the rotor speed values, which may be sensibly different during transients.

Ref. [21] proposes a data-driven method that makes use of ambient PMU measurements to estimate the inertia constant of all individual generators. The results are accurate, and the methodology also provides the center of inertia frequency. However, the estimation strategy is based on a sliding window of 300 s which needs to be carried out before updating the inertia values, posing a limitation for online applications.

A dynamic optimal forgetting factor recursive least square inertia estimator is proposed in [22]. During the system identification process, a transfer function downgrade to the first order is required. Therefore, the method is not able to carry out the estimation of devices that include other controls with different dynamic order than the inertial response, such as the PFC. Furthermore, it makes use of electrical frequencies measured at the generation bus, and this may lead to estimation errors considering power imbalance events since this measurement can differ from the internal frequency of the device.

A method that is able to estimate the inertia and other parameters of SGs and CIGs by using ambient measurements is proposed in [23]. The power system is modeled using stochastic differential equations and, when the system variables oscillate around a steady-state equilibrium point, the covariance matrix of bus voltage and line current measurements can be used to extract the inertia constants by solving a least-square problem. Reported results are accurate and the method is able to track time-varying inertia. However, there is no clear indication regarding the location of the measurements to be taken. Furthermore, the effect of measurement noise is not considered. Therefore, there is no experimental proof regarding the robustness of the method to imperfect measurements.

Ref. [24] provides a data-driven method for estimating the parameters of SGs and CIGs using ambient measurements. The SG parameters can be extracted from the sensitivity matrix that relates active power variations from rotor angle variations, which can be estimated by applying the least-square method if enough measurements are available. Even though reported results are generally accurate, the methodology requires rotor angles and rotor speed measurements, which are often not available to the TSO.

An algorithm capable of estimating the internal frequency of an SG is proposed in [25]. It makes use of a Thevenin equivalent circuit of the SG and employs PMU measurements taken at the generation bus. Once the internal frequency of the device has been computed, an adaptive unscented Kalman filter is used to carry out inertia estimations. However, the method is applicable only to SGs and is not able to extract meaningful information from ambient measurements.

An ARMAX-based method for estimating the inertia of SGs is introduced in [26]. The method provides accurate inertia estimates and is robust to measurement noise and outliers. However, the algorithm requires an active power perturbation to occur since it is not able to carry out the inertia estimation under normal operating conditions. Furthermore, it requires the instant of the disturbance to be known to isolate the inertial response of the SG from other dynamics (such as PFC) since it is not able to discern the different power contributions. In this regard, it is worth highlighting that this may pose a limitation in power systems where there is a significant presence of CIGs that provide fast PFC that overlaps with the inertial response.

This article proposes a system identification approach for online parameter estimation of SG and CIG technologies. The proposed method has an element-wise scope; i.e., it is applied separately at each SG or CIG. Remarkably, it is able to carry out accurate estimates when the power system is under normal operating conditions, and after a power imbalance event. The methodology requires a certain knowledge of the

device being monitored. Particularly, it requires knowing the number of poles and zeros of the corresponding s-domain transfer function. However, as detailed later in Section 2, this can be easily deduced by knowing the type of device and its associated control.

Contribution

This paper proposes a novel inertia estimator based on system identification methods. Its main advantages over other works are the following:

- Apart from the inertia constant, the proposed method allows the estimation of other useful parameters of the device to be monitored, such as the PFC droop gain, and the damping coefficient.
- It is able to carry out the estimation of the inertia and other parameters using both ambient measurements (normal operating conditions) and ringdown signal measurements (power imbalance events).
- It is currently one of the few estimators applicable to CIGs. Furthermore, it does not require knowledge of the power deviation of the event that triggers the estimation.
- The methodology is suitable for online applications since it presents very low computational times on the order of 10^{-4} s. To the best of our knowledge, no other online inertia estimator exhibits lower running times.
- It merely requires measurements proceeding from a PMU installed at the generation bus. This is a reasonable assumption since NERC standards recommend the presence of a PMU at the interconnection point for plant modeling [27] and operation and control [28]. Therefore, it is applicable in real-life implementations.
- It does not depend on the size of the network since it is applied in a decentralized way, i.e., the system identification of every device is carried out independently.

Paper organization

The rest of the paper is organized as follows. Section 2 gives a brief description of the algorithm used in this work, namely, the recursive least-squares equation error algorithm, and the SG and CIG models in their s-domain and z-domain forms. In Section 3, the proposed methodology is presented. Section 4 introduces the case study and the simulation results on the IEEE 39-bus and IEEE 118-bus test systems. Finally, Section 5 presents the conclusions of this study.

2. Description of the algorithms

2.1. Recursive least-squares equation error model

The recursive least-squares (RLS) equation error (EE) is an adaptive infinite input response (IIR) filter and is characterized by presenting the possibility to adapt its transfer function parameters during the estimation process. Due to this property, IIR filters present superior system modeling capabilities compared to finite impulse response (FIR) filters. A brief description of the RLSEE is presented below, but the interested reader can refer to [29] for additional details and find information about other adaptive versions, such as the least mean square or the lattice versions.

In the EE formulation, the information vector is composed of current and past input and output measurements and is described by the following expression:

$$\boldsymbol{\Phi}(k) = [d(k-1) \ \dots \ d(k-n_d) \ x(k) \ x(k-1) \ \dots \ x(k-n_x)]^T, \tag{1}$$

where k is the iteration index, vector d represents the measured output signal, x is the input signal vector, and n_d and n_x are the adaptive filter denominator and numerator orders, respectively.

The parameter vector, whose elements correspond to the IIR filter discrete-time domain transfer function parameters, is given by:

$$\boldsymbol{\theta}(k) = [-a_1(k) \ \dots \ -a_{n_d}(k) \ b_0(k) \ b_1(k) \ \dots \ b_{n_x}(k)]^T, \tag{2}$$

where $a_1, \dots, a_{nd}$ and $b_0, \dots, b_{nx}$ are the denominator and numerator parameters of the z-domain transfer function, respectively. The equation error is defined as:

$$e(k) = d(k) - \boldsymbol{\theta}(k)^T \boldsymbol{\Phi}(k). \tag{3}$$

Adopting the following least-squares function as the objective function to be minimized:

$$\xi(k) = \sum_{i=0}^{k} \lambda^{k-i} e(i)^2 = \sum_{i=0}^{k} \lambda^{k-i} [d(i) - \boldsymbol{\theta}(k)^T \boldsymbol{\Phi}(i)]^2, \tag{4}$$

it is possible to derive the RLS algorithm, which has the objective of tuning the adaptive filter parameters $\boldsymbol{\theta}(k)$ such that the computed output signal $y(k) = \boldsymbol{\theta}(k)^T \boldsymbol{\Phi}(k)$ matches the measured output signal d during the observation period. The RLS repeats, at each iteration k, the basic following steps:

$$e(k) = d(k) - y(k) \tag{5}$$

$$\boldsymbol{S}_{De}(k+1) = \frac{1}{\lambda} \left[\boldsymbol{S}_{De}(k) - \frac{\boldsymbol{S}_{De}(k) \boldsymbol{\Phi}(k)^T \boldsymbol{\Phi}(k) \boldsymbol{S}_{De}(k)}{\lambda + \boldsymbol{\Phi}(k)^T \boldsymbol{S}_{De}(k) \boldsymbol{\Phi}(k)} \right] \tag{6}$$

$$\boldsymbol{\theta}(k+1) = \boldsymbol{\theta}(k) + \boldsymbol{S}_{De}(k+1) \boldsymbol{\Phi}(k) e(k), \tag{7}$$

where $\boldsymbol{S}_{De}$ is the coefficient-error-vector covariance matrix of $(n_d + n_x, n_d + n_x)$ dimensions, and λ is the forgetting factor parameter used to discount the weight of distant past information when the coefficient is updated. λ is typically chosen in the range $0 < \lambda < 1$. Matrix $\boldsymbol{S}_{De}$ is usually initialized with a diagonal matrix with large number diagonal values (e.g., 10^5 or higher) [30–32].

In a given iteration k, the transfer function of the EE formulation is expressed by the following general representation in the discrete-time domain z:

$$H_k(z) = z^{n_d - n_x} \cdot \frac{b_0(k) z^{n_x} + b_1(k) z^{n_x - 1} + \dots + b_{n_x - 1}(k) z + b_{n_x}(k)}{z^{n_d} + a_1(k) z^{n_d - 1} + \dots + a_{n_d - 1}(k) z + a_{n_d}(k)}. \tag{8}$$

Fig. 1 shows the basic configuration for system identification applications of the equation error formulation. Functions $A(k, z)$ and $B(k, z)$ are the polynomial representations of the parameters previously introduced and are defined as follows:

$$A(k, z) = b_0(k) + b_1(t) z + \dots + b_{n_x}(k) z^{n_x} \tag{9}$$

$$B(k, z) = 1 + a_1(k) z + \dots + a_{n_d}(k) z^{n_d}. \tag{10}$$

Compared with FIR filters, IIR filters exhibit lower computational complexity and are therefore more suited for online applications. Furthermore, IIR filters have higher system modeling capabilities because of their adapting poles. Certainly, IIR filters have some drawbacks, such as possible instability, the presence of several local minima in the error surface, and a slower convergence [29]. The EE formulation avoids these disadvantages because it has only a global minimum [29]. This is a consequence of the output $y(t)$ being a linear combination of the filter parameters. Additionally, the EE formulation presents good convergence properties if it is properly initialized.

2.2. SG and CIG transfer function and their z-domain equivalent

First, consider the well-known differential equation that represents the dynamics of the SG rotor speeds [33]:

$$2H_g \frac{d\Delta\omega_g}{dt} = \Delta P_{PFC} - \Delta P_g - D\Delta\omega_g, \tag{11}$$

where H_g is the SG inertia constant, $\Delta\omega_g$ represents the SG rotor speed variations, ΔP_{PFC} is the variation of the mechanical power provided by

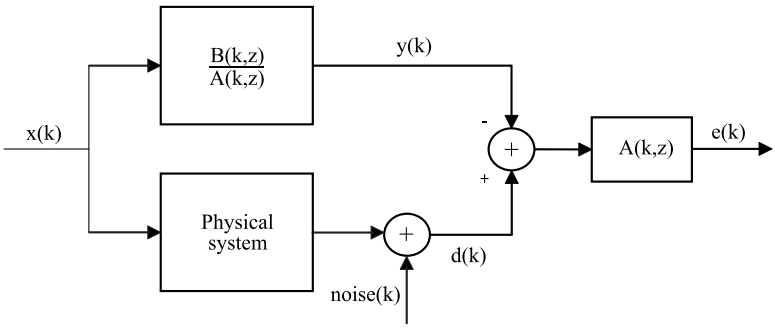


Fig. 1. Equation error formulation for system identification applications.

the turbine due to PFC, ΔP_g represents the variation of the electromagnetic power supplied by the generator, and D is the machine damping coefficient. Taking the Laplace transform of (11), one can derive the s-domain transfer function of SG with PFC:

$$\frac{\Delta\omega_g(s)}{\Delta P_g(s)} = -\frac{s\frac{1}{Mg} + \frac{1}{TM_g}}{s^2 + s(\frac{1}{T} + \frac{D}{M_g}) + \frac{D+1/R}{TM_g}}, \tag{12}$$

where $M_g = 2H_g$ is the mechanical time constant of the SG. In the previous expression, the PFC is represented with a simplified model, taking into account the droop gain and a single time delay block as follows:

$$\Delta P_{PFC} = -\frac{1/R}{1+Ts}\Delta\omega_g, \tag{13}$$

where R is the droop gain and T is an equivalent time constant of the PFC system that accounts for the governor and turbine dynamics. Clearly, with this simplification, the proposed methodology will not be able to detect internal governor and turbine dynamics. However, in this study, we are interested solely in determining the PFC droop gain. Taking the same steps, it is possible to derive the transfer function of an SG neglecting the PFC:

$$\frac{\Delta\omega_g(s)}{\Delta P_g(s)} = -\frac{\frac{1}{M_g}}{s + \frac{D}{M_g}}. \tag{14}$$

This expression will be used later in Section 4 to estimate the parameters of a generator representing an equivalent network.

Considering a general CIG with a virtual synchronous machine (SVM) grid-forming control scheme [34] and using the same approach as before, one has the following s-domain transfer function:

$$\frac{\Delta\omega_{CIG}(s)}{\Delta P_{CIG}(s)} = -\frac{\frac{1}{M_{CIG}}}{s + \frac{G}{M_{CIG}}}, \tag{15}$$

where G represents the inverse of the droop gain and M_{CIG} is the virtual mechanical time constant provided by the CIG. In this study, if not specified otherwise, all the aforementioned magnitudes are expressed in per unit (p.u.), apart from M_g and M_{CIG}, which are expressed in seconds. It is worth highlighting that other control schemes for CIG are currently available in the literature, such as the virtual synchronous generator and the synchronverter. However, in this work, we only focus on the VSM since it is one of the most common controls associated with CIGs. The interested reader can find a detailed overview of these control schemes, along with their advantages and inconveniences, in [34,35].

It is worth mentioning that models represented in (12) and in (15) do not take into account the internal dynamics of the generation unit and of the power electronic converter. However, these dynamics are not important for the estimation of M and PFC droop gain, which is the main focus of this work. Furthermore, as described in [36,37], using a system identification model with an order higher than necessary may lead to poor estimations since the measurement noise may be interpreted as a higher-order dynamic of the system. This is particularly true for the EE formulation, where the error is directly included in the model and the algorithm tries to minimize its contribution as it pertains to the unknown system model. In this work, the order of the model has been determined empirically. However, Ref. [37] recently proposed an optimal order selection for system identification models that may be used for this purpose.

At this point, the relationship between the s-domain transfer functions previously described and their z-domain equivalent is used [38], which can be obtained through the aforementioned RLSEE algorithm. Having in mind that the following s-domain transfer functions have these corresponding z-domain counterparts:

$$F(s) = \frac{1}{s+a} \quad ; \quad F(z) = \frac{z}{z - e^{-aT_s}} \tag{16}$$

$$F(s) = \frac{sf+g}{(s+c)^2+h^2}; \quad F(z) = \frac{z^2 f - z[fe^{-cT_s}cos(hT_s) - \frac{g-cf}{h}sin(hT_s)]}{z^2 - 2ze^{-cT_s}cos(hT_s) + e^{-cT_s}}, \tag{17}$$

where T_s is the sampling period, it is possible to determine the relationship between the parameters determined using the RLSEE and the s-domain transfer function parameters. Comparing Eqs. (14) and (15) with (16) and the general z-domain transfer function form (8), one obtains:

$$M_{CIG} = -\frac{1}{b_0}T_s \quad ; \quad G = \frac{-ln(a_1)}{T_s}M_{CIG}T_s \tag{18}$$

$$M_g = -\frac{1}{b_0}T_s \quad ; \quad D = \frac{-ln(a_1)}{T_s}M_gT_s, \tag{19}$$

where b_0 and a_1 are the coefficients determined using the RLSEE formulation. It is worth remarking that (18) and (19) are valid for CIGs and SGs without an associated PFC, respectively. Similarly, comparing (12) with (17) and the general z-domain transfer function form (8), it is possible to determine the parameter of an SG with PFC. After mathematical manipulations, one has:

$$M_g = -\frac{1}{b_0}T_s \quad ; \quad T = \frac{-T_s}{ln(a_2)}; \quad 1/R = \left(h^2 + \frac{1}{4T^2}\right)TM_g, \tag{20}$$

where a_2 is provided by the RLSEE algorithm and h is computed using the following expression $h = \arccos(-a_1/2\exp(-T_s/2T))/T_s$ [38].

2.3. Model independent estimator of generator rotor speeds

As described in the previous section, to carry out the system identification and estimate the generation unit parameters we need the internal frequency variation signal. In practice, it may not be accessible to the TSO. However, it can be estimated using a recently introduced model-independent rotor speed estimator [39] based on the frequency

divider formula, in which a convex optimization problem is solved to derive the following formula:

$$\Delta\boldsymbol{\omega}_g = (\boldsymbol{D}^T\boldsymbol{D})^{-1}\boldsymbol{D}^T\Delta\boldsymbol{\omega}_B = \boldsymbol{D}^+\Delta\boldsymbol{\omega}_B, \tag{21}$$

where $\Delta\boldsymbol{\omega}_g$ is the vector of the generator internal frequencies to be estimated, $\Delta\boldsymbol{\omega}_B$ is the vector of the bus frequency measurements, $\boldsymbol{D}$ is the frequency divider matrix and $\boldsymbol{D}^+$ is its left inverse. One of the main advantages of this formulation is that each generator rotor speed can be estimated independently from the others. Therefore, it is compatible with the element-wise approach of this article. In fact, the $\boldsymbol{D}^+$ matrix is quite sparse, and the non-null elements of its rows correspond to the bus frequency measurements required to estimate the internal frequency of the generator associated with the analyzed row. It is worth pointing out that in real electric networks, most of the power plants are connected to the rest of the system with an antenna connection by means of a substation. In this case, the only non-null elements of the $\boldsymbol{D}^+$ row are related to the generation bus and its only neighboring bus, i.e., the low-voltage bus and the high-voltage bus of the elevating transformer of the substation. Therefore, this method only requires the installation of a PMU in the substation, which is a common practice in recent years and recommended by NERC standards, as previously mentioned.

3. Method description

As shown in Fig. 2, the proposed methodology makes use of PMUs connected at the generation buses. Particularly, this instrumentation provides active power measurements and bus frequency measurements.

Signal preprocessing is important to improve the system identification accuracy [36]. Therefore, all the measurements are converted into per unit and are detrended. Furthermore, since the measurement noise can jeopardize the estimation accuracy, it is important to filter it out. In this study, we employ a simple low-pass filter for both active power and bus frequency measurements.

As shown in the s-domain transfer functions of the previous section, the system identification method needs active power variations (as input signals) and rotor speed variations (as output signals). As previously mentioned, the latter is estimated using the model-independent estimator [39], whereas the former is directly measured.

Once the shape of the transfer function in the s-domain is known, it is possible to determine the orders n_d and n_x of the EE formulation. Particularly, an SG with PFC represented by the s-domain transfer function indicated in Eq. (12) has an EE formulation with $n_d = 2$ and $n_x = 2$. However, the s-domain formulations of the SGs without PFC (14) and of CIGs (15) have z-domain counterparts characterized by having $n_d = 1$ and $n_x = 1$. It is important to keep this in mind when employing the EE formulation for parameter estimation.

Once the measurements have been filtered and the rotor speed variations (or the internal frequency variations, in the case of CIGs) have been determined, these measurements are used to compute the z-domain transfer function parameters applying the RLSEE algorithm. Subsequently, Eqs. (18), (19), and (20) are used to obtain the s-domain transfer function parameters, which have a more tangible physical meaning. In this study, we focus on the estimation of the mechanical time constant M (which is closely related to the inertia constant) and the PFC droop gain. It has to be noted that (12) only takes into account the electromechanical oscillations and the PFC of the machine. Therefore, it does not consider faster dynamics such as the ones of damping windings, which are in any case included in the models used by the simulator, as detailed later. In this sense, it is worth highlighting that since these dynamics act on different time frames, the approximation taken in (12) allows an accurate estimation of the inertia constant and the PFC droop gain.

Finally, it is worth mentioning that the only information required by the proposed RLSEE estimator are the input and output signals, and the order of the EE formulation, which is determined by the shape of the transfer function of the device under consideration. The proposed method, apart from the order of the transfer function, is model-independent and can be applied to any device without having any knowledge about its parameters.

4. Test results

In this section, test results on modified versions of the IEEE 39-bus and of 118-bus test systems are reported. To prove the effectiveness of the proposed method with non-synchronous devices, several SGs have been substituted with CIGs. All of these CIG devices employ a VSM grid-forming control scheme. The plant modeling and the control blocks have been implemented in a DIgSILENT PowerFactory environment using DSL language in accordance with work [40]. These control functions include virtual inertia and active power control, reactive power control, virtual impedance, and phase locked loop. The SGs are modeled in PowerFactory using the so-called called model 2.2, i.e., a round rotor model with one damping winding on the d-axis and two damping windings on the q-axis. Each SG has an associated IEEE-G1 governor and IEEE-T1 exciter. The measurements of frequency and active power at the generation bus used to carry out estimations are extracted from the software DIgSILENT PowerFactory assuming the availability of a PMU. With this respect, it is worth remarking that the presence of a PMU at a generation bus providing the required measurements is not only realistic but also recommended by NERC standards [27,28]. If not specified otherwise, in all of the simulations reported in this section, random Gaussian noise as described in [41] is added to the measurements proceeding from the PMUs. The sampling time used for input and output signals is 10 ms, which can be easily achieved by commercially available PMUs [17]. It should be noted that the proposed method is not affected by the sampling rate of the measurements, as long as it is sufficiently fast to capture the dynamics of interest. All measurements are filtered using the MATLAB *lowpass* function with a cutoff frequency of 0.5 Hz, which has been determined heuristically by carrying out several simulations. In fact, it is worth mentioning that all the generation units in both the IEEE 39-bus and IEEE 118-bus test systems do not present significant dynamics with higher frequency components. Therefore, by using this cutoff frequency value, the effect of measurement noise is reduced without losing any significant information for carrying out the parameter estimation. Every simulation is initialized considering the system at steady-steady conditions with a nominal frequency of 60 Hz at every bus. Afterwards, an event that provokes a power imbalance in the network occurs at $t = 1$ s. The proposed algorithm has been coded in MATLAB. The computer used to carry out the simulations is an Intel Core i7-1255U 1.70 GHz with 16 GB of RAM.

4.1. Test results on the IEEE 39-bus test system

In this subsection, several simulation results are provided to show the accuracy of the proposed algorithm. The simulation is conducted considering the modified IEEE 39-bus test system represented in Fig. 3. The generator connected to bus 39 represents a neighboring system of the U.S. and Canada. The synchronous generators connected to buses 30 and 38 in the original system have been substituted with a wind farm and a photovoltaic (PV) plant, respectively. Both of these plants have a nominal power of 1000 MVA and initially produce an active power of 250 MW and 830 MW, respectively. A relatively small electrochemical storage system of 30 MVA connected to bus 34 is also considered. This device does not supply any active power during the initial steady-state conditions since it is supposed to provide only emulated inertia and PFC services. Each SG has an associated IEEE-G1 governor and IEEE-T1 exciter.

In all of the following simulations, the RLSEE algorithm makes use of a forgetting factor of $\lambda = 1 - 10^{-4}$. The covariance matrix S_{De} is initialized with a diagonal matrix in which the diagonal values

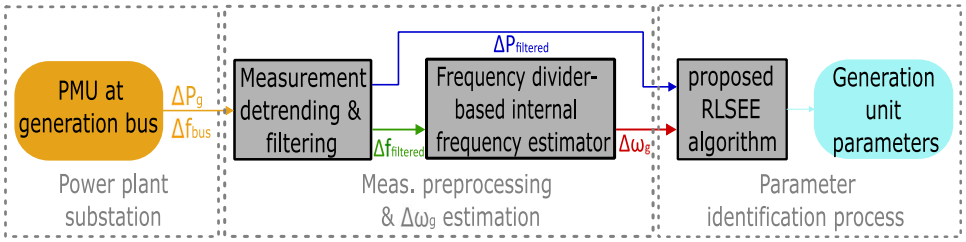


Fig. 2. Flowchart of the proposed methodology.

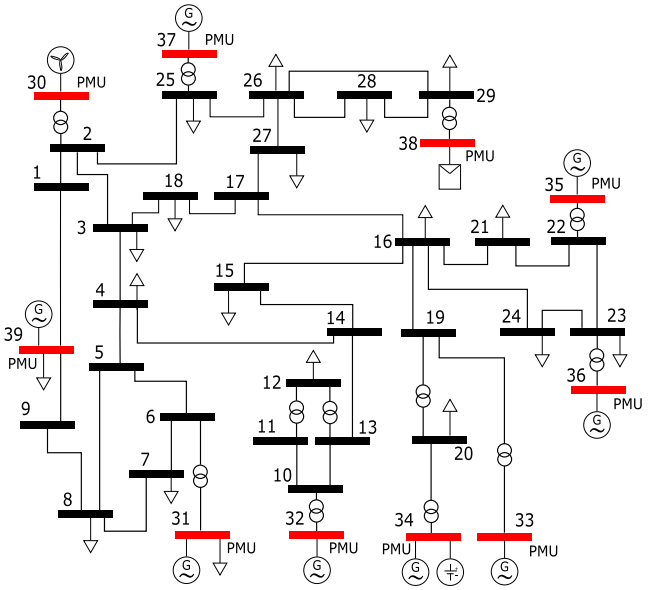


Fig. 3. Single-line diagram with PMU locations of the IEEE 39-bus test system.

correspond to 10^{10} in the case of the SG connected to bus 39, to 10^5 in the case of the wind farm and the PV plant, and to 10^6 in the case of the battery system. The parameters of the RLSEE algorithm are initialized with null values. A proper initialization of the RLSEE algorithm is required to achieve good convergence properties [29].

4.1.1. Case 1.1: Several power imbalance events

In this case study, we present the accuracy of the proposed algorithm considering several events. These events take place in different parts of the test network and consist of load outages, increases and decreases in load power consumption, and synchronous generator outages. The steady-state values of mechanical time constant M estimations are reported in Table 1. These values are referred to as 100 MVA, which is taken as the reference power of the system.

The first two events consider the sudden outage of load 27 and load 3, which determines an active power imbalance of 281 MW and 322 MW, respectively. The reported error is quite small for every generation unit, with only one estimation error above 4%. Since these events occur in a relatively central part of the network and all the SGs and CIGs are mostly affected equally by these occurrences, the M estimation error is quite similar among generation units. The average absolute error is rather small, being below 2% in both cases.

Other simulations assume loads 15 and 7 suddenly increase their consumed active power by a factor of 50%, resulting in a power imbalance of −160 MW and −116.9 MW, respectively. The last load event assumes load 29 decreases its active power consumption by 50%, involving a power imbalance of 141.75 MW. These events imply a sensibly smaller power deviation compared with the first two cases. These smaller perturbations are responsible for higher average absolute errors compared with the first two load events. Furthermore, there are more pronounced differences between the individual M estimations among the generation units. This can be explained because loads 7 and 29 are located in a peripheral part of the network, and therefore, there will be a significant difference in how these events affect the generation units.

In the last two cases, we assume the power plants connected to buses 36 and 37 reduce their delivered active power by 50%. This could be caused by an internal fault or an untimely trip of an internal protection device that disconnects half of the plant generators. The active power imbalance of these events is −280 MW and −270 MW, respectively. The M estimation of the affected generator is not reliable with the proposed method since the sudden power decrease is interpreted as an inertia contribution. However, since the plant is affected by a contingency, estimating its internal parameters is of little interest. The last two columns of Table 1 show that the average absolute error for these events is rather small and that the M estimation of every SG and CIG is accurate.

Although the aforementioned events occur in different parts of the system and involve different kinds of perturbations with different active power deviations, the proposed algorithm provides accurate results. In fact, the estimation errors of individual generation units are always below 7%, and the average estimation error is always below 4% in all considered events.

As previously mentioned, we observe that the nearer the perturbation that provokes a power imbalance is to a generation unit, the better tends to be its M estimation. Similarly, the farther the incident, the worse the M estimation. This can be explained by taking into account that nearer power imbalances provoke higher electromechanical oscillations of SGs and CIGs. Consequently, the higher the electromechanical oscillations are, the better the proposed algorithm can extract the dynamic response of the device, thus estimating its transfer function parameters more accurately. In fact, certain adaptive filtering methods, such as the EE formulation, need involved signals to be sufficiently exciting [29].

The simulation that assumes an increase of 50% of the consumed active power from load 7 is the one with the highest average errors. This makes sense since the event occurs at a relatively peripheral point of the system, and the involved power deviation is the lowest among the considered events. In contrast, events such as the two load outages and the two generation events, which involve higher power imbalances or occur in more central parts of the network, allow the proposed algorithm to provide better estimations with lower average errors.

Fig. 4 shows the estimated values of the discrete time transfer function parameters of generator 36 (considering the generator 37 contingency) with the RLSEE. These values are used to conduct the M and PFC droop gain estimations by means of Eq. (20) previously provided. At this point, it is worth reminding that SGs with PFC have a 4th-order EE formulation with $n_d = 2$ and $n_x = 2$, i.e., coefficients a_1, a_2, b_0, and b_1 have to be estimated. To obtain a comparison point, we conducted a parameter estimation with an iterative EE. With this method, the obtained parameter values are $a_1 = -1.992$; $a_2 = 0.9916$; $b_0 = -1.965 \cdot 10^{-4}$; and $b_1 = 1.057 \cdot 10^{-4}$. The steady-state values obtained with the RLSEE and represented in Fig. 4 are $a_1 = -1.987$;

Table 1
Case study 1.1 — Steady-state estimation errors of M.

		Load 27 contingency	Load 3 contingency	Load 15 contingency	Load 7 contingency	Load 29 contingency	SG 36 contingency	SG 37 contingency
	Actual M (MWs/MVA)	M estim. % error	M estim. % error	M estim. % error	M estim. % error	M estim. % error	M estim. % error	M estim. % error
Wind farm	80	−1.47%	0.88%	−1.83%	−5.91%	5.37%	−3.57%	2.19%
PV plant	50	−1.86%	0.87%	−1.35%	−4.80%	−2.39%	−0.35%	−1.49%
BESS	35	−2.06%	−2.06%	−3.65%	0.00%	−0.36%	0.47%	1.38%
SG 31	60.6	0.29%	1.85%	6.21%	−1.69%	−0.97%	−1.69%	−1.43%
SG 32	71.6	2.10%	−0.53%	−4.35%	−5.46%	−6.07%	1.24%	−3.17%
SG 33	57.2	−0.02%	2.23%	−1.35%	−5.92%	−3.06%	−2.63%	−4.05%
SG 34	52	−4.28%	−2.33%	−3.55%	−5.95%	−5.48%	−1.37%	6.32%
SG 35	69.6	2.11%	1.55%	0.67%	−4.99%	−5.05%	−2.66%	0.36%
SG 36	52.8	−1.73%	0.33%	1.60%	−0.91%	2.57%	–	−1.94%
SG 37	48.6	−3.55%	1.80%	−4.23%	6.72%	0.95%	2.45%	–
SG 39	1000	0.94%	0.72%	−1.38%	−1.88%	0.11%	1.45%	−0.04%
Average error	–	1.86%	1.38%	2.74%	4.02%	2.94%	1.79%	2.24%

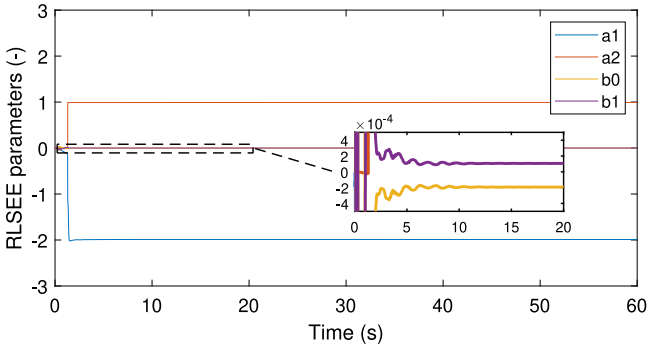


Fig. 4. Case 1.1 — Generator 36 RLSEE parameters (Gen. 37 contingency).

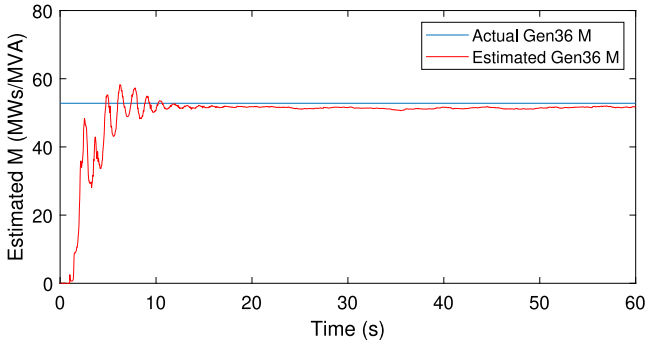


Fig. 5. Case 1.1 — Generator 36 M estimation (Gen. 37 contingency).

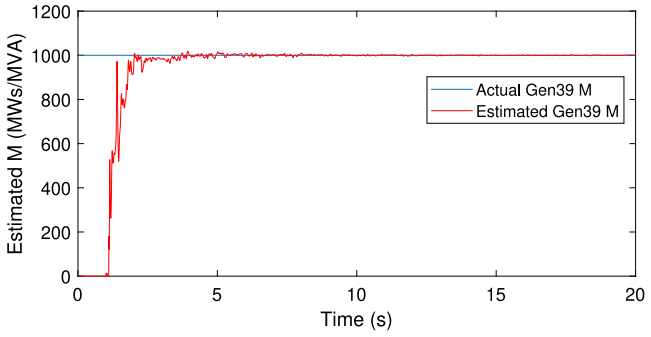


Fig. 6. Case 1.1 — Generator 39 M estimation (Gen. 37 contingency).

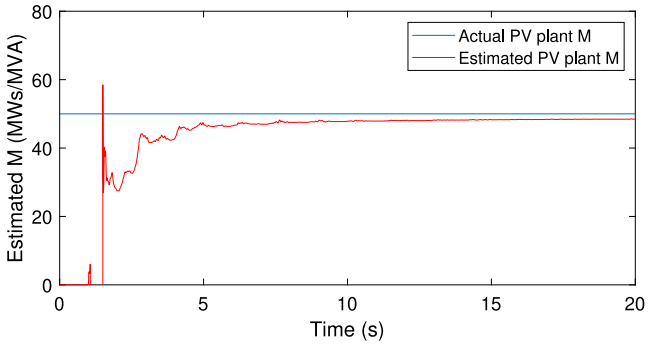


Fig. 7. Case 1.1 — PV plant M estimation (Gen. 37 contingency).

$a_2 = 0.9913$; $b_0 = -1.932 \cdot 10^{-4}$; and $b_1 = 1.061 \cdot 10^{-4}$. Since the parameters provided by the recursive algorithm are very similar to those of its iterative counterpart, we conclude that the RLSEE algorithm is correctly implemented [29].

Using the parameters of the z-domain transfer function shown in Fig. 4 and applying the equations of Section 2, computing M and the PFC droop gain is straightforward. Fig. 5 shows the M estimation of generator 36. After a few seconds, the M estimation approaches the actual value of 52.8 MWs/MVA, and after some further oscillations, it reaches a steady-state value that is very similar to the true M value. Similar considerations can be made for other SGs.

Fig. 6 shows the estimated M values of generator 39 (which represents an equivalent network) after the generator 37 contingency. The figure shows that the estimation is precise and that after approximately 2 s, the value provided by the proposed methodology approaches the actual generator inertia of 1000 MWs/MVA. After $t = 3$ s, the estimate oscillates to reach the steady-state value of 999.6 MWs/MVA.

The M estimates of the PV plant are represented in Fig. 7. The figure shows that the proposed methodology is also applicable to CIG plant parameter estimations. Compared with the estimation of SG 36, the estimates do not exhibit the same oscillations prior to reaching the steady-state value. This can be explained because the oscillations of the internal frequency of the CIGs tend to be more damped due to their control settings. Since the proposed methodology extracts the parameter information based on the measured electromechanical response of the device, the M estimation of CIGs reaches the steady-state value with smaller oscillations compared to SGs.

Table 2 shows the estimation errors of the parameters $1/R$ and G, which represent the inverse of the PFC droop gain of the synchronous

Table 2
Case study 1.1 — Steady-state estimation errors of 1/R and G.

		Load 27 contingency	Load 3 contingency	Load 15 contingency	Load 7 contingency	Load 29 contingency	SG 36 contingency	SG 37 contingency
	Actual 1/R-G values (MW/MVA)	1/R-G % error	1/R-G % error	1/R-G error	1/R-G % error	1/R-G % error	1/R-G % error	1/R-G % error
Wind farm	70	0.96%	0.82%	1.11%	1.00%	0.78%	1.20%	1.06%
PV plant	50	0.43%	−2.67%	0.52%	−0.43%	−1.35%	0.62%	1.08%
BESS	35	−0.93%	−0.92%	−0.93%	−0.92%	−0.92%	−0.93%	−0.94%
SG 31	29.75	−0.26%	−0.12%	0.27%	0.60%	−0.25%	0.08%	0.14%
SG 32	34	−0.30%	−0.01%	0.03%	0.64%	−0.02%	0.03%	0.02%
SG 33	34	−0.30%	−0.06%	0.15%	0.46%	−0.13%	−0.10%	0.08%
SG 34	25.5	0.04%	0.03%	−0.06%	−0.06%	0.06%	−0.15%	−0.03%
SG 35	34	0.04%	−0.13%	0.27%	−0.22%	−0.27%	0.15%	0.14%
SG 36	29.75	0.04%	−0.04%	0.09%	−0.44%	−0.08%	–	0.05%
SG 37	29.75	0.04%	0.22%	−0.47%	0.06%	0.45%	−0.06%	–
SG 39	–	–	–	–	–	–	–	–
Average error	–	0.33%	0.50%	0.38%	0.48%	0.43%	0.37%	0.39%

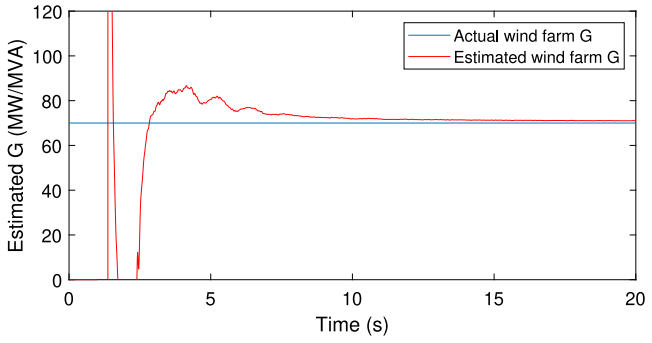


Fig. 8. Case 1.1 — Wind farm G estimation (Gen. 37 contingency).

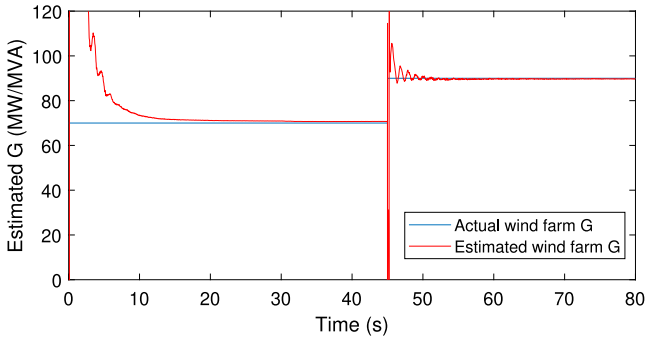


Fig. 9. Case 1.2 — Wind farm G estimation (Load 27 contingency).

generators and the CIGs, respectively. Similar to the M estimates, these values also refer to the reference power of the system. In every simulation, the inverse of the PFC droop gain is estimated with a very small percentage error, with an average absolute error well below 1%. Fig. 8 shows that after a few seconds the proposed method provides an accurate estimate of the droop gain. As previously mentioned, this is particularly relevant in the case of CIGs, since their PFC droop constant is not known *a priori*, as it can be changed by acting on the related plant control scheme.

It can be observed that the convergence time of the proposed method can differ from case to case. For instance, the M estimation of SG 36 depicted in Fig. 5 needs approximately 10 s to stabilize, whereas a lower time is needed in the case of SG 39, the PV power plant, and the wind farm. This can be explained by noticing that the considered event is the outage of SG 37, which is electrically far from SG 36. Thus, the input and output signals used for its parameter estimations are not as effective as the ones used for the other generation units. Therefore, coherently with Ref. [29], the RLSEE convergence time for SG 36 is higher. From Figs. 7 and 8, it is also possible to notice that, immediately after the contingency, the estimations present significant oscillations (which are not completely depicted to better capture the more relevant part of the estimate). In fact, the PV plant and the wind farm are electrically near the SG 37 contingency and the frequency measurements at their generation bus suffer from numerical problems, typical of PMUs, during discontinuous events [39]. Therefore, the RLSEE applied to these generators presents higher oscillations compared to the ones of SG 36 and SG 39.

4.1.2. Case 1.2: Change in the droop gain of CIGs

An important property that the proposed estimator must possess is the ability to detect sudden changes in the PFC droop gain of CIGs, since this variable is controllable by acting on the control system. In fact, one of the main challenges for TSOs is the ability to verify that

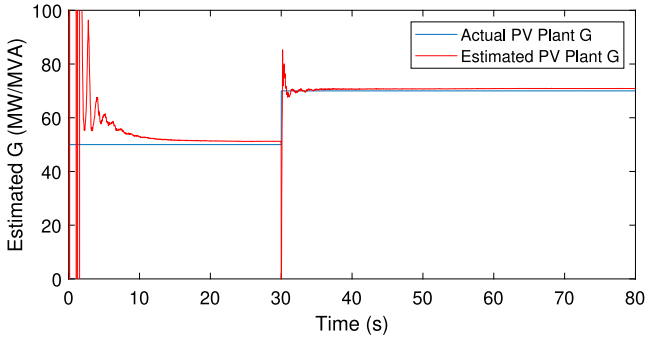


Fig. 10. Case 1.2 — PV plant G estimation (Load 27 contingency).

CIGs are actually providing the required services [1]. To demonstrate this capability, the G parameter of both the wind farm and the PV plant is changed by acting on their control system implemented in PowerFactory. The simulated scenario is the trip of load 27 at $t = 1$ s previously described. To compensate for the frequency variation, the G of the PV plant is supposed to increase its PFC contribution from 50 to 70 MW/MVA at $t = 30$ s, whereas the G of the wind farm increases from 70 to 90 MW/MVA at $t = 45$ s. Figs. 9 and 10 show that the proposed methodology is capable of promptly detecting changes in the parameters, thus allowing the TSO to have a precise picture of the actual frequency contribution of CIGs.

4.1.3. Case 1.3: Influence of measurement noise

This section reports a case study in which measurements are affected by noise with a doubled maximum value compared with the previous study cases. The considered noise is still characterized by having a Gaussian distribution and a zero mean value, as indicated in [41].

Table 3
Case study 1.3 — Steady-state estimation errors of M and 1/R-G with different noise values (load 27 contingency).

	Normal noise	Doubled noise	Normal noise	Doubled noise
	M estim. % error	M estim. % error	1/R-G estim. % error	1/R-G estim. % error
Wind farm	−1.47%	−0.46%	0.96%	0.96%
PV plant	−1.86%	3.19%	0.43%	0.49%
BESS	−2.06%	0.05%	−0.93%	−0.96%
SG 31	0.29%	0.22%	−0.26%	−0.52%
SG 32	2.10%	−5.91%	−0.30%	−0.55%
SG 33	−0.02%	3.97%	−0.30%	−0.40%
SG 34	−4.28%	−4.06%	0.04%	0.06%
SG 35	2.11%	0.31%	0.04%	0.18%
SG 36	−1.73%	−5.67%	0.04%	0.37%
SG 37	−3.55%	2.53%	0.04%	−0.05%
SG 39	0.94%	−0.10%	–	–
Av. error	1.86%	2.41%	0.33%	0.45%

Table 3 shows the estimation errors of both the M and PFC droop gain parameters. The considered event is the load 27 contingency, but similar results have been encountered with other perturbations. It can be observed that the presence of more pronounced noise does not significantly affect the M estimation, since no individual generation unit has an estimation error higher than 6%. Furthermore, the average absolute error increases from 1.86% to 2.41%, which remains an acceptable value. The estimation of the PFC droop gain is also slightly affected by the presence of a stronger noise, but their percentage error remains very small. Thus, the proposed algorithm appears to be robust against noise levels that are sensibly higher compared to normal values.

4.2. Test results on the IEEE 118-bus test system

This section analyzes the accuracy of the estimation provided by the proposed method when applied to the IEEE 118-bus test system. The power flow data and the topology configuration of the network can be found in [42]. Since the dynamic data of the system is not available, it has been supposed by assigning a technology to each power plant (hydroelectric, nuclear, etc.) coherent with its rated apparent power, and equipping each SG with a governor, an automatic voltage regulator, and a power system stabilizer. Since the estimation of the M and PFC droop gain is particularly interesting for CIGs, this subsection focuses on these devices, whereas less attention is given to SGs since their M and droop gains are known constant parameters. As in the previous study case, several SGs have been substituted with CIGs of the same apparent power with a VSM control scheme; their location, along with their M and PFC droop gain constants, are detailed in Table 4. Overall, the system is composed of 19 generation units, 10 SGs, and 9 CIGs. The RLSEE algorithm makes use of a forgetting factor $\lambda = 1 - 10^{-4}$ and is initialized supposing a diagonal covariance matrix S_{De} with diagonal elements of 10^{10}.

In this network, the power consumption of loads is modeled using an Ornstein–Uhlenbeck (OU) stochastic process with a standard deviation of 1% and an autocorrelation coefficient of 0.5 1/s [43]. By doing so, the system will present stochastic active power and frequency oscillations typical of real power systems under normal operating conditions. In this manner, it is possible to analyze the capability of the proposed method to extract meaningful estimations by using ambient measurements.

4.2.1. Case 2.1: Several power imbalance events

The proposed method is tested by simulating several events that cause power and frequency oscillations throughout the network. Similarly to the simulations carried out in the IEEE 39-bus system, the simulations last 20 s and the event occur at time $t = 1$ s. It is important to highlight that, throughout the whole simulation time, the stochastic process of loads is considered. The power imbalance caused by the events, along with the steady-state relative errors of the M and G estimations, are reported in Tables 5 and 6, respectively. It is possible to observe that the events are located in different parts of the system and involve different power imbalances, ranging from 12 MW to 184 MW. The active power sign related to each load contingency depends on the type of event: load outages involve a positive power imbalance, whereas a load surge (case of load 90) involves a negative power imbalance. Additionally, a line outage is considered to verify whether this type of incidents can be used by the proposed method to carry out parameter estimation.

It is possible to observe that the proposed method is able to accurately extract the inertia value of all the CIGs, with an accuracy that improves when the power imbalance reaches several tens of MW. Interestingly, the proposed method presents quite accurate estimations also considering small power imbalances of less than 20 MW. In this regard, it is worth mentioning that the RLSEE algorithm benefits from the presence of the OU stochastic process associated with load power consumption. In fact, the power and frequency oscillations provoked by the stochastic noise of loads contain useful information for parameter estimation and allow accurate estimates, as analyzed in the next study case. This is an interesting feature of the proposed method since real power systems under normal operating conditions are subject to this kind of stochastic process [43] and they can be actually exploited by the RLSEE estimator.

Table 4
IEEE-118 bus system — Connection bus, M, and G constants of CIGs.

Bus	49	54	59	61	65	66	80	89	100
Actual M (MWs/MVA)	25	9	30	13	60	80	30	98	49
Actual G (MW/MVA)	25	7.5	25	18.2	70	60	42	56	35

Table 5
Case study 2.1 — Steady-state estimation relative errors of M.

CIG at bus	49	54	59	61	65	66	80	89	100
Load 116 outage (184 MW)	0.3%	3.4%	1.8%	−2.0%	−2.4%	−3.4%	−4.4%	−3.8%	0.1%
Load 90 +50% (−81 MW)	−0.7%	−4.1%	2.3%	1.6%	−3.6%	−2.9%	1.7%	0.0%	5.7%
Load 70 outage (66 MW)	1.3%	4.1%	1.7%	−1.0%	−1.6%	−2.2%	−2.0%	−1.3%	−1.5%
Load 7 outage (19 MW)	3.1%	5.4%	0.3%	−2.2%	−3.8%	−4.3%	−7.3%	−0.8%	4.7%
Load 58 outage (12 MW)	4.9%	1.2%	−0.3%	−2.0%	−2.7%	−7.2%	−7.9%	1.2%	2.0%
Line 63–64 outage	0.5%	1.1%	3.4%	2.2%	−2.8%	−2.9%	−2.4%	0.2%	1.3%

Table 6
Case study 2.1 — Steady-state estimation relative errors of G.

CIG at bus	49	54	59	61	65	66	80	89	100
Load 116 outage (184 MW)	0.5%	0.7%	1.5%	−2.3%	−0.8%	−2.3%	−5.0%	−3.6%	1.9%
Load 90 +50% (−81 MW)	0.5%	0.3%	1.6%	−2.4%	−0.4%	−2.6%	−4.1%	−0.1%	2.6%
Load 70 outage (66 MW)	0.5%	0.8%	1.8%	−2.5%	−0.9%	−2.1%	−4.7%	−3.5%	2.2%
Load 7 outage (19 MW)	0.7%	1.5%	2.9%	−3.0%	−1.5%	−1.7%	−5.1%	−4.4%	3.0%
Load 58 outage (12 MW)	1.0%	4.0%	4.7%	−3.4%	−2.1%	−1.1%	−5.4%	−5.0%	3.8%
Line 63–64 outage	1.0%	2.0%	−6.1%	−3.1%	−0.9%	−4.5%	−5.7%	−3.4%	0.7%

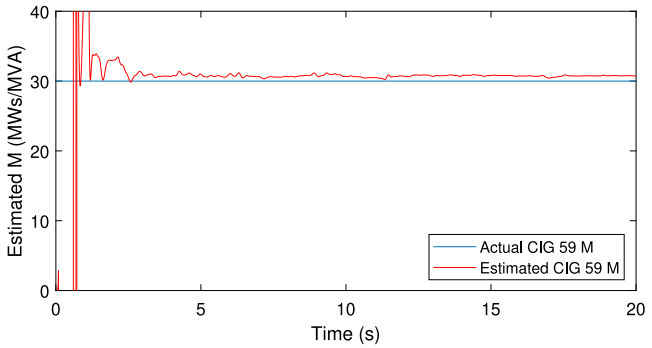


Fig. 11. Case 2.1 — CIG 59 M estimation (load 90 contingency).

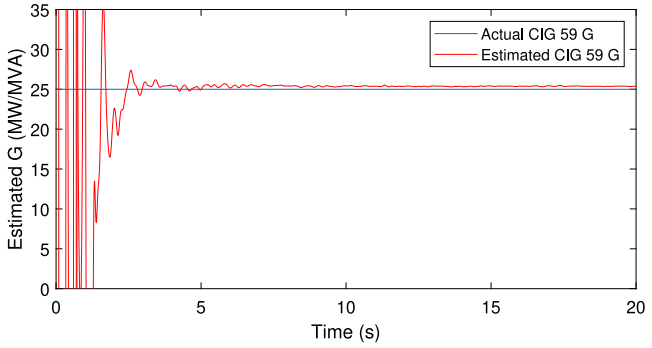


Fig. 12. Case 2.1 — CIG 59 G estimation (load 90 contingency).

The M and the G estimations of CIG 59 after the load 90 contingency are reported in Figs. 11 and 12, respectively. The estimates approach the actual value 1–2 s after the event and they stabilize around simulation time $t = 4$ s. The parameter estimation of the other devices, for all the simulated events, presents a similar behavior.

4.2.2. Case 2.2: Normal operating conditions and time-varying M and G of CIGs

In this study case, the ability of the proposed RLSEE algorithm to carry out parameter estimation when the power system is under normal operating conditions is analyzed. Therefore, no power imbalance event is simulated and only stochastic fluctuations of loads are considered. Interestingly, the proposed algorithm is also able to track time-varying values of M and G by using only ambient measurements. For this purpose, a 20-min simulation is considered. During this time-lapse, CIGs connected to buses 54, 61, and 66 are supposed to change their virtual M. Particularly, their initial M values are the ones of Table 4. At simulation time $t = 360$ s, CIG 54 increases its virtual M by 50%, and CIGs 61 and 66 by 100%. Subsequently, at $t = 780$ s, CIG 54 and CIG 66 re-establish their initial M value, whereas CIG 61 presents a final M value of 19.5 MWs/MVA. Furthermore, CIGs connected at buses 59 and 65 change their PFC droop gain G. At simulation time $t = 480$ s, CIG 59 doubles its G, and CIG 65 reduces its PFC droop gain by 33%. Afterwards, at $t = 720$ s, CIG 65 restores its initial G value, whereas CIG 59 presents a final G value of 43.75 MW/MVA. These changes are depicted in Fig. 13 and in Fig. 14.

In the literature, it is known that the RLS algorithm has the drawback of losing adaptation alertness after a significant simulation time. This is due to the fact that, over time, the eigenvalues of the covariance matrix S_{De} become small, reducing the tracking capability of the algorithm [29]. Since the simulation of this study case lasts 20 min, this effect becomes non-negligible and results in a slow tracking of the time-varying M and G parameters. In order to avoid this problem, a simple covariance matrix resetting algorithm is employed [44]. Particularly,

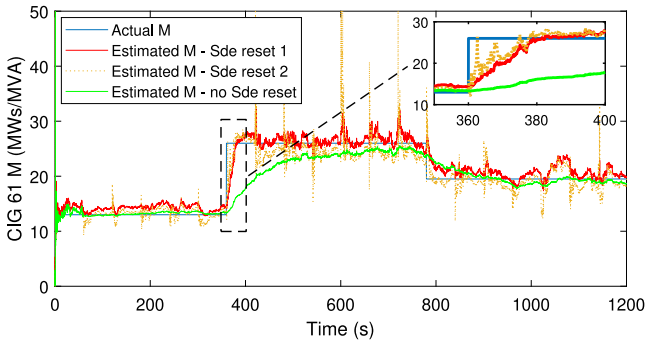


Fig. 13. Case 2.2 — Estimation of time-varying M of CIG 61 using ambient measurements.

for all the CIGs, the covariance matrix S_{De} is reset every 60 s using the following diagonal matrix:

$$S_{De_{reset1}} = \begin{bmatrix} 10^5 & 0 \\ 0 & 10^2 \end{bmatrix}, \tag{22}$$

which has been determined empirically. It should be noted that, in the case of CIGs, the S_{De} matrix has $(2, 2)$ dimensions since $nd = nx = 1$, and that the variance terms of 10^5 and of 10^2 are representative of the estimation uncertainties of parameters a_1 and b_0, respectively. Empirically, it has been observed that assigning higher uncertainties to a_1 leads to better adaptation alertness, and that the covariances (off-diagonal terms) do not significantly affect the reactivity of the algorithm.

In Fig. 13, the M estimation of CIG 61 with different covariance reset matrices is reported. The estimations using the covariance reset matrix $S_{De_{reset1}}$ are depicted in red color. It is possible to observe that with this setting the RLSEE is able to carry out a reliable estimation of M and to track its M in about 20 s after the occurrence (t = 360 s). The estimations present some oscillations around the actual value and are not always as precise as in the case of power imbalance events, but they can be used from the TSO to detect M changes without waiting for a particular event to occur. The estimations obtained without resetting the covariance matrix are depicted in green. Also in this case, the RLSEE algorithm is able to detect time-varying M but with lower precision and slower tracking time since it has smaller adaptation alertness.

The dashed orange curve represents the estimates using the following reset covariance matrix:

$$S_{De_{reset2}} = \begin{bmatrix} 10^5 & 0 \\ 0 & 10^3 \end{bmatrix}. \tag{23}$$

With this setting, the RLSEE is able to detect M and G changes but it presents significant oscillations throughout the simulation and notable spikes when the covariance matrix is reset, i.e., every 60 s. Therefore, a careful reset of the covariance matrix should be carried out. In the literature, methods that employ exponential or cyclic resetting of the S_{De} are proposed [45], and they may be integrated with the proposed RLSEE to automatically determine the covariance matrix resetting. However, this matter will be addressed in a future work.

Fig. 14 represents the M estimates of CIGs 54 and 56, along with the G estimations of CIGs 59 and 65; all of them are obtained using $S_{De_{reset1}}$. The proposed algorithm is able to provide reliable estimations of M and G and to track their changes in 20–40 s, depending on the case. Similarly to the M estimation of CIG 61, the estimates present some oscillations and are less precise compared to the case where power imbalance events are considered. However, the RLSEE proves to be able to yield reasonable M and G estimates, and, with a proper covariance matrix reset, to detect their changes over time. Since the presence of CIGs is going to increase in most power systems, and their inertia and PFC droop gains may be time-varying parameters based

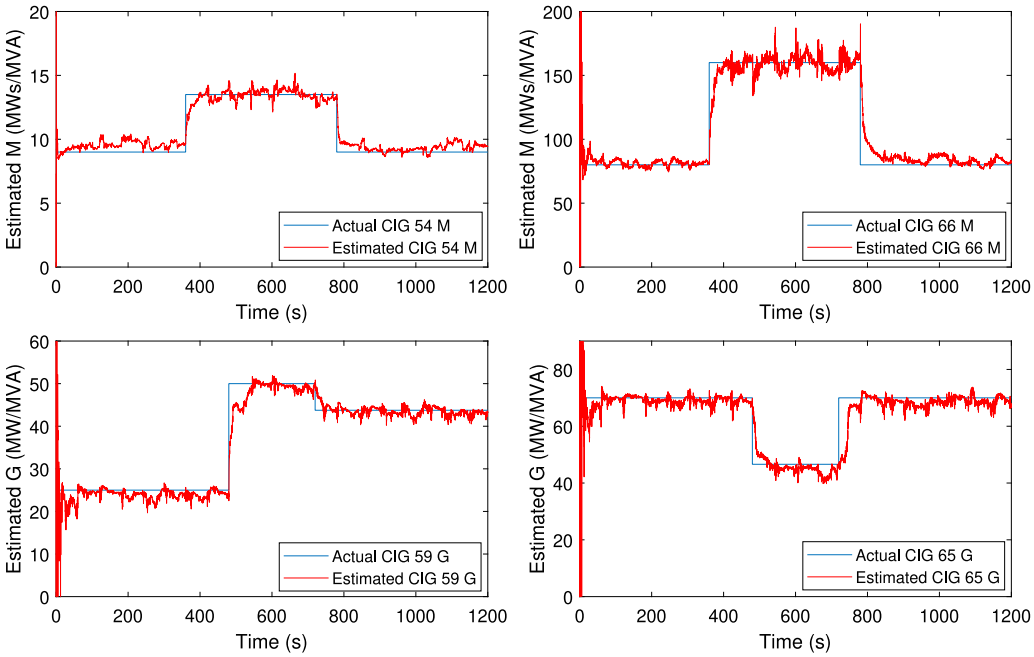


Fig. 14. Case 2.2 — Estimation of time-varying M and G of different CIGs using ambient measurements.

on the instructions received from the TSO, this RLSEE feature seems particularly interesting. Furthermore, the estimation accuracy may be further increased by adopting advanced adaptive methods [45].

It is worth remarking that the changes of the M constants and of the PFC droop gain occur at different moments. Thus, the estimation is carried out by considering only ambient measurements since the frequency and active power oscillations generated by the G changes cannot be used by the RLSEE to carry out the M estimation, and viceversa.

4.3. Further remarks and comparison with other works

In all of the presented study cases, the parameters have been initialized with zero values, i.e., with the TSO having no information about the devices to be monitored. However, in a real-life scenario the operator may use an educated guess or a previously estimated value, which would reduce the convergence time of the proposed methodology.

Remarkably, the proposed methodology exhibits very low computational times. In fact, each iteration is carried out in approximately $3 \cdot 10^{-4}$ s in the case of SGs and in approximately $5 \cdot 10^{-5}$ s in the case of CIGs. The difference between these two values is due mainly to the necessity of adopting a higher-order model with 4 parameters in the case of SGs instead of the second-order model used for estimating the CIG parameters. These computational times are well below the supposed PMU sampling time of 10 ms, and therefore, the methodology is suitable for online applications.

It is worth mentioning that the measurements of electrical frequency taken at generation buses from PMUs may be affected by numerical problems, especially during the first instants after a nearby contingency. This may affect the internal frequency estimation using [39] and, in turn, influence the parameter estimation. However, since the proposed methodology employs a forgetting factor that gives increasingly smaller weight to the estimates as the simulation goes on, the biased estimates of the first instants after the contingency do not affect the accuracy of the method, but only the convergence time.

Table 7 illustrates a comparison with other elementwise online inertia estimation work in terms of computational burden, average estimation error, and scope. The table shows that the methodology proposed in this article exhibits the lowest computational burden. Regarding the estimation accuracy, it can be noticed that the proposed method is comparable with other online inertia estimation methods with an elementwise scope. However, it is worth noting that most of the other methods allow only the estimation of the SG inertia time constant, whereas our methodology permits estimating also other parameters and is applicable for both SGs and CIGs.

Table 7
Comparison with other works.

Reference	Computational time (s)	Average error of M	Device	Parameter
[12]	$\approx 10^{1}-10^{3}$	≈2%	SG	M
[13]	Not reported	≈1%	SG	M
[16]	Not reported	≈1–2%[a]	SG/CIG	M/G
[18]	$\approx 10^{0}$	≈1–2%	SG/CIG	M
[19]	$\approx 10^{-1}$	≈15%	SG	M
[20]	$\approx 10^{0}$	≈2–3%	SG	M
[21]	$\approx 10^{-1}$	≈3–4%	SG/CIG	M
[23]	Not reported	≈4–5%	SG/CIG	M/G
[24]	$\approx 10^{-1}$	≈1–4%	SG/CIG	M
[25]	$\approx 10^{-1}$	≈0.5–1%	SG	M
[26]	$\approx 10^{-1}$	≈1–2%	SG	M
This work	$\approx 10^{-4}$	≈2–3%	SG/CIG	$M/G/D/T$

[a]The average value has been extrapolated from the figures of the estimation trajectories.

For the sake of completeness, we provide a numerical comparison with one of the aforementioned inertia estimators, namely, the inertia calculation application (ICA) proposed in [13]. This algorithm first computes the rate of change of frequency (RoCoF) by measuring the electrical frequency at the generation bus and then, by assuming that the power imbalance of an event is known, it carries out the inertia estimation of a single device. This estimate is accurate during the first instants immediately after the perturbation. Subsequently, when the PFC activates, the method is not able to provide a reliable estimation since the active power variation does not depend solely on the inertial response and the method does not have the capability to discern the two contributions. However, there are cases in which the estimation of SG inertia is always compromised. For example, if there is a significant

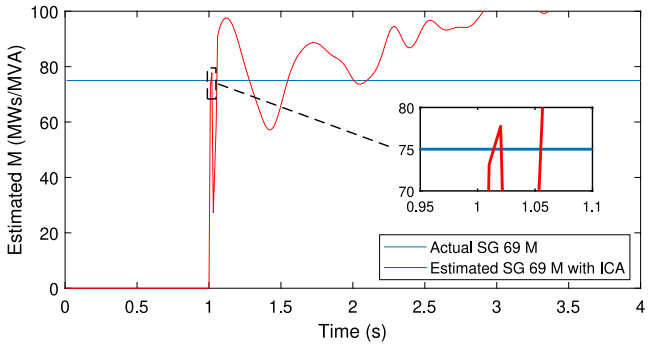


Fig. 15. Case 3 — SG 69 M estimation without fast PFC (IEEE 118-bus system load 70 contingency).

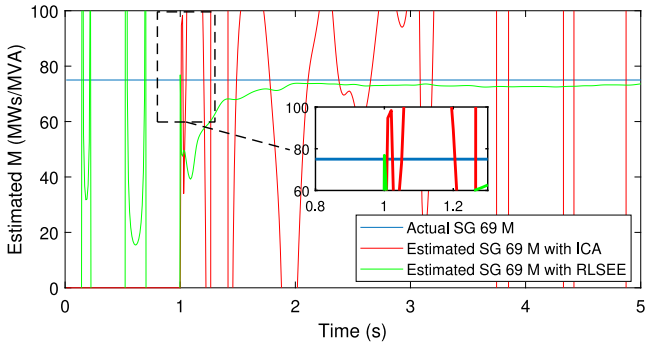


Fig. 16. Case 3 — SG 69 M estimation with fast PFC (IEEE 118-bus system load 70 contingency).

amount of CIGs in the network that provides fast PFC service, like in our modified IEEE 118-bus system, there is an overlap between the inertial response and the fast PFC that compromises the accuracy of the ICA method.

To support these considerations, we run two simulations on the modified IEEE 118-bus system considering the load 70 outage detailed in Table 5. The M estimation of the SG connected to bus 69 is considered. In the first simulation, the fast PFC of all the CIGs is disabled. Observing Fig. 15, it is possible to notice that the estimation provided by the ICA method is accurate, although only immediately after the perturbation.

In the second simulation, the fast PFC of CIGs is activated and the estimation provided by the ICA is not reliable anymore. As can be noticed by observing Fig. 16, the M estimation provided by the ICA is sensibly higher than the actual one. On the contrary, the proposed RLSEE algorithm is able to accurately track the M estimation because it is able to discern the active power variation caused by the inertial response and the one provoked by the PFC, since they act on different dynamic orders. In the author's mind, these simulation results show an important limitation of RoCoF-based estimators since power systems are likely to experience a growing presence of CIGs in future years.

5. Conclusion

This study proposes a new methodology to estimate the transfer function parameters of SGs and CIGs, including the mechanical time constant M and the PFC droop gain. The proposed RLSEE algorithm is a measurement-based estimator capable of providing accurate estimates after large disturbances and under normal operating conditions of the power system, i.e., by using ambient measurements.

Simulation results on the IEEE 39- and 118-bus test systems confirm the high accuracy of the proposed estimator. Moreover, the small computational times of about 10^{-4} s make it suitable for real-time monitoring and control applications. The algorithm is also shown to be robust against high noise levels, maintaining good estimation accuracy for both the mechanical time constant and the droop gain estimations. It is worth reminding that the proposed method is independent of the size of the network since the estimation is carried out with a decentralized approach. In fact, the estimation of each device's parameters merely relies on the PMU measurements at the generation bus.

Considering power imbalance events, the algorithm performs better and shows faster convergence properties with events that cause significant power imbalances. Generally, the nearer the events are to a device, the better its parameter values can be extracted. However, events with smaller power imbalances can also be used by the proposed methodology to carry out the parameter estimation, showing its feasibility for real-life power system operation.

When the system is under normal operating conditions, the RLSEE algorithm provides reliable estimates with small convergence times. Moreover, by applying a proper covariance matrix reset, the proposed method is also capable of tracking time-varying M and G parameters, which is one of the main challenges that power electronics devices pose to TSOs.

In future research, the proposed methodology may be applied to CIGs with different control strategies, such as the virtual synchronous generator and the synchronverter control schemes. Additionally, future work may be devoted to the implementation of advanced resetting techniques of the covariance matrix to improve the tracking capability of the RLSEE estimator. Another research line may consist of estimating controllable load parameters to allow the TSO to evaluate their actual behavior in real-time. Finally, a covariance resetting version of the RLSEE may be implemented to speed up the estimation under events with small active power imbalances.

CRediT authorship contribution statement

Davide Gotti: Conceptualization, Methodology, Investigation, Software, Validation, Formal Analysis, Writing – original draft. **Pablo Ledesma:** Supervision, Writing – review & editing. **Hortensia Amaris:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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